LESSON 14.5
CALCULATING THEORETICAL PROBABILITY OF COMPOUND EVENTS

LESSON INTRODUCTION

MA.8.DP.2.2 Find the theoretical probability of an event related to a repeated experiment.

LEARNING TARGETS

- I can calculate theoretical probability of compound events.

BENCHMARK COHERENCE

BUILDING ON

MA.7.DP.2.3 Find the theoretical probability of an event related to a simple experiment.

WORKING TOWARD

MA.912.DP.4.3 Calculate the conditional probability of two events and interpret the result in terms of its context.

ESSENTIAL QUESTIONS

- How are simple events different from compound events?
- How do you find the theoretical probability of compound events?

LESSON NARRATIVE

In this lesson, students build on their understanding of calculating theoretical probability. Students begin by creating a game using cards, a spinner, or a 12-sided die and determining the probability of winning the game. Students extend their understanding of probability to compound probability, exploring how to calculate the probability of two independent compound events.

COMPONENT SUMMARY

14.5.1 WARM-UP *(~5 MIN.)*

Order probabilities and connect to a game

14.5.2 ACTIVITY *(10-15 MIN.)*

Create a game given simple experiments

14.5.3 GUIDED PRACTICE *(10 MIN.)*

Select and calculate probability of compound events

14.5.4 TRY IT *(10 MIN.)*

Calculate probability of independent events

14.5.5 YOUR TURN! *(5-10 MIN.)*

Calculate probability of independent events from real-world examples

14.5.6 WRAP-UP *(~5 MIN.)*

Self-reflection on the content of the lesson

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- Independent events

MATERIALS

- (Optional) Scientific calculators

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may confuse one event with multiple outcomes as being more than one event.
- Students may try to add to find the probability of independent events instead of multiplying.

THINGS TO CONSIDER

- In order for students to complete the 14.5.2 Activity in the allotted time, students may need guidance. Suggestions are included in that component below.
- Students will find the probability with replacement during the 14.5.4 Try It! Consider exposing students to multiple examples that involve replacement in order to assist students with understanding how to calculate probability with replacement.

LITERACY AND LANGUAGE ROUTINES

Clarify, Critique, Correct

The 14.5.2 Activity provides an opportunity for students to use the Clarify, Critique, Correct routine. For question 7, encourage students to critique each other's games and to ask clarifying questions. For question 8, they correct any errors found in the games.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.3.1 Mathematical Fluency

As students continue to engage with probability questions throughout this unit, they are building mathematical fluency with probability. They are offered multiple opportunities to practice efficient and applicable methods. Monitor the ways students are using feedback from the teacher and peers to improve efficiency and accuracy when calculating probabilities.

DIFFERENTIATE INSTRUCTION

Many components throughout this lesson allow for differentiated instruction through student choice. The 14.5.1 Warm-Up allows students to choose from five given numbers and to use it, as a decimal, fraction, or percentage, to represent probability of a game. The 14.5.2 Activity allows students to choose simple experiments to create a game, and the 14.5.3 Guided Practice allows students to choose probabilities to practice finding the probability of a compound event.

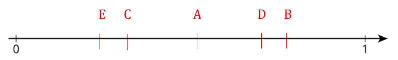
Additional differentiation opportunities are denoted throughout the Lesson Guide.

14.5.1 WARM-UP: ORDERING PROBABILITIES

14.5.1 Warm-Up: Ordering Probabilities

1. Locate and label these numbers on the number line.

   A. 0.5
   B. 0.75
   C. 0.33
   D. 0.67
   E. 0.25

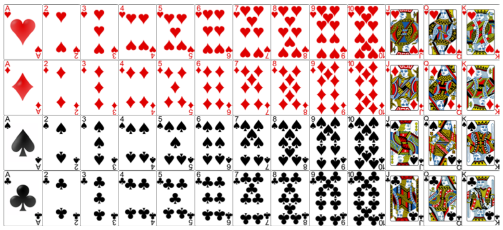


2. Choose one of the numbers from the previous question. Describe a game in which that number represents your probability of winning.

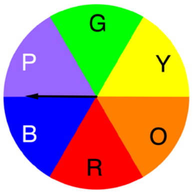
   Answers will vary. Possible answer describing point A could be the chances of landing on heads when flipping a two sided coin.

14.5.2 Activity: Make Your Own Game

Option 1:
A card is drawn from a standard deck of 52 playing cards. There are 52 cards in a standard deck of cards; half are red (hearts and diamonds), half are black (spades and clubs), and 12 are "face" cards (jacks, queens, and kings). Aces are considered to have a value of one.



Option 2:
The fair spinner, shown, is spun.



Option 3:
A fair 12-sided die is rolled.

Work with your partner to complete the following.

Step 1: Choose any two simple experiments from the options provided.

Step 2: Create a game with your partner using the two options you chose.

1. Describe the game you created.
   Answers will vary.
2. Determine the number of outcomes in the sample space.
   Answers will vary. This will depend on the game chosen.
3. Define Event A = the outcomes resulting in a win.

   Event A = *Answers will vary. This will depend on the game chosen.*
4. How many outcomes are in Event A?
   Answers will vary. This will depend on the game chosen.
5. List the outcomes in Event A.
   Answers will vary. This will depend on the game chosen.
6. What is the probability of winning your game?
   Answers will vary. This will depend on the game chosen.
7. Switch games with another partner pair. Verify their work.
8. Make any adjustments necessary to your game based on the other partner pair's review of your game.

14.5.2 ACTIVITY: MAKE YOUR OWN GAME

TEACHER MOVES

In order to clearly communicate expectations, provide an example of a game previously created in a prior lesson or other example. Preview and discuss key components of the activity prior to beginning so students understand what they will be doing and have the opportunity to ask clarifying questions before getting started.

DIFFERENTIATE INSTRUCTION

For question 1, consider providing a chart to complete in order for students to organize and record the elements of the game they created. For example:

Game Title: Experiments Chosen Number of Players How to Win Rules

!!!

14.5.3 GUIDED PRACTICE: CALCULATING PROBABILITY

DIFFERENTIATE INSTRUCTION

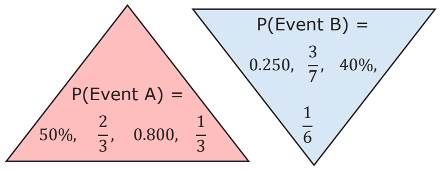
Allow informal observation throughout the first two components to dictate how "guided" this Guided Practice should be for you and your classroom. This could mean small-group instruction for some while others attempt independently, or it could mean teacher facilitation of practice with students actively working as a whole group.

QUESTIONING

For question 1, encourage students to include a real-world example that represents the probability that they chose from each triangle. For example: P("Event A") = $\frac{1}{3}$; the probability of choosing to wear black pants, blue pants, or khaki colored plants to school. P("Event B") = 0.250; the probability of randomly selecting the current season out of a bag that contains all four seasons.

14.5.3 Guided Practice: Calculating Probability

1. Choose a probability of Event A from the P(A) triangle. Choose a probability of Event B from the P(B) triangle. Use those probabilities to calculate P(A and B). In your work, use P notation to specify which probability you are using: P(A), P(B), and P(A and B).



Answers will vary.
P(A)= 0.800, P(B)= 0.250, and P(A and B)=0.200
Or
P(A)= 0.800, P(B)= $\frac{3}{7}$, and P(A and B)=$\frac{12}{35}$
Or
P(A)= 0.800, P(B)= 40%, and P(A and B)=0.32
Or
P(A)= 0.800, P(B)= $\frac{1}{6}$, and P(A and B)=$\frac{2}{15}$
Or
P(A)= 50%, P(B)= 0.250, and P(A and B)=0.125
Or
P(A)= 50%, P(B)= $\frac{3}{7}$, and P(A and B)=$\frac{3}{14}$
Or
P(A)= 50%, P(B)= 40%, and P(A and B)=0.2
Or
P(A)= 50%, P(B)= $\frac{1}{6}$, and P(A and B)=$\frac{1}{12}$
Or
P(A)= $\frac{2}{3}$, P(B)= 0.250, and P(A and B)=$\frac{1}{6}$
Or
P(A)= $\frac{2}{3}$, P(B)= $\frac{3}{7}$, and P(A and B)=$\frac{2}{7}$
Or
P(A)= $\frac{2}{3}$, P(B)= 40%, and P(A and B)=$\frac{4}{15}$
Or
P(A)= $\frac{2}{3}$, P(B)= $\frac{1}{6}$, and P(A and B)=$\frac{1}{9}$
Or
P(A)= $\frac{1}{3}$, P(B)= 0.250, and P(A and B)=$\frac{1}{12}$
Or
P(A)= $\frac{1}{3}$, P(B)= $\frac{3}{7}$, and P(A and B)=$\frac{1}{7}$
Or
P(A)= $\frac{1}{3}$, P(B)= 40%, and P(A and B)=$\frac{2}{15}$
Or
P(A)= $\frac{1}{3}$, P(B)= $\frac{1}{6}$, and P(A and B)=$\frac{1}{18}$

2. Suppose you have an evenly divided spinner that shows all the days of the week. You also have a bag of papers that list the months of the year.
Suppose you spin the spinner and pick a paper from the bag.

What is the theoretical probability that you choose the paper with the current month and the spinner lands on the current day of the week? Use P notation when calculating the probability.
P(current day of the week)=$\frac{1}{7}$, P(current month)=$\frac{1}{12}$, P (current day and month)= $\frac{1}{84}$. The theoretical probability of landing on the correct day of the week and the correct month is $\frac{1}{84}$, or about 1.2%

14.5.4 Try It: Calculating Probability

1. As part of a game at the school carnival, the player picks a marble from a bag, replaces the marble, and picks a marble from the bag again. The bag of marbles has two green, two red, and two blue marbles.
   a. Calculate the theoretical probability that a player chooses a green marble and a blue marble. Write the probability as a fraction, as a decimal rounded to the nearest thousandth, and as a percent rounded to the nearest tenth of a percent.
   $P(green\ and\ blue) = \frac{1}{9}\ or\ 0.111\ or\ 11.1\%$
   b. Calculate the theoretical probability that a player doesn't pick a red on either of their picks.
   $P(no\ red) = \frac{4}{9}\ or\ 0.444\ or\ 44.4\%$

2. An integer chip with one yellow side and one red side is tossed 15 times.
   a. What is the probability that the first toss results in red and all subsequent tosses result in yellow?
   $0.5^{15} = 0.0000305\ or\ 0.00305\%$
   b. What is the probability that all 15 tosses result in the same color?
   $0.5^{15} = 0.0000305\ or\ 0.00305\%$

14.5.4 TRY IT: CALCULATING PROBABILITY

TEACHER MOVES

The key word "replace" in question 1 is essential to finding the compound probability. Consider having a discussion about how the key word "replace" affects the probability. Students may comment that the process to find compound probability involves only multiplication since replacing the first marble does not change the probability of the second marble. Students may also list synonyms for "replace" that have the same effect on probability, such as "put back" or "return."

ENRICHMENT

In order for students to interpret the meaning of the probability found in question 2, ask questions to foster understanding such as:

- Why are the answers for questions 2a and 2b the same answer?
- For question 2b, would the probability of all of the chips landing on red be the same probability as all of the chips landing on yellow? Why or why not?

14.5.5 YOUR TURN!

ENRICHMENT

For question 1b, assist any struggling students with understanding and finding the probability of the first event by asking questions such as:

- What is the probability of getting a dog, as one event?
- What is the probability of getting a cat, as one event?
- What is the probability of getting a dog or a cat, as one event?
- What is the first event?

14.5.5 Your Turn!

1. Although she would like to have more, Wanda is only allowed to have two pets total. She is allowed to choose from these three options: a dog, a cat, and a rabbit. Wanda has decided to use a die to randomly choose which two pets she will have. If she rolls a 1 or a 2, she will get a dog, if the die lands on a 3 or 4 she will get a cat, and she will get a rabbit if the die lands on 5 or 6.

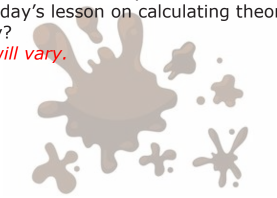
   Calculate the theoretical probability that Wanda will end up with two cats.
   $\left(\frac{2}{6}\right)\left(\frac{2}{6}\right) = 0.\bar{1} \text{ or } 11.\bar{1}\%$

2. A textbook has exactly 428 pages numbered in order starting with 1 and ending with 428. You flip to a random page in the book in a way that it is equally likely to stop at any of the pages. Then you repeat the process.

   Consider Event N, turning to an odd numbered page three times in a row.
   Calculate the theoretical probability of Event N.
   $Event\ N = (0.5)(0.5)(0.5) = 0.125 \text{ or } 12.5\%$

14.5.6 Wrap-Up: Reflection

1. What was the ***muddiest*** (most difficult or confusing) point in today's lesson on calculating theoretical probability?
   Answers will vary.



14.5.6 WRAP-UP: REFLECTION

DIFFERENTIATE INSTRUCTION

The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.

TEACHER MOVES

Consider pairing students to discuss their reflections to see if peers can answer any of their questions.

